

Sandun Induwara Herath

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PROFILE

Biomedical engineering undergraduate with research experience in computer vision, machine learning and biomedical signal processing. My primary research interests lie in developing impactful research solutions on computer vision domain.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [J.1] Pamuditha Somarathne, **Sandun Herath**, Gaetano Gargiulo, Paul Breen, Neil Anderson, Yu Yao, Tongliang Liu, Anusha Withana. **A Dual Classifier-Regressor Architecture for Heart Sound Onset/Offset Detection.** *IEEE Transactions on Biomedical Engineering* (January 15, 2026). <https://doi.org/10.1109/TBME.2026.3654558>.

EXPERIENCE

- **University of Moratuwa** Jun 2026 - Present
Instructor Moratuwa, Sri Lanka
 - Conduct laboratory sessions and tutorials for undergraduate engineering courses.
 - Mentor students in managing projects and about signal processing concepts.
- **The University of Sydney** Dec 2024 - Jul 2025
Visiting Research Student NSW, Australia
 - Developed machine learning and algorithmic approaches for automated heart sound segmentation.
 - Designed annotation software and signal processing pipelines for PCG-ECG analysis.

EDUCATION

- **University of Moratuwa** 2022 - 2026
BSc. Engineering (Hons.) in Biomedical Engineering Moratuwa, Sri Lanka
 - cGPA: 3.86/4.00
 - Relevant Courses: Deep Learning for Vision, Image Processing and Computer Vision, Advances in Vision, Digital Signal Processing, Medical Image Processing, Optimization
- **Central College Anuradhapura** 2012 - 2020
GCE Advanced Level - Physical Science Stream Anuradhapura, Sri Lanka
 - 3A passes with zscore 2.6277

PROJECTS

- **Instructional Guided Digital Histopathology** Jun 2025 - Ongoing
Final year project
 - Parameter efficient fine-tuning of vision-language foundation models for medical report generation from histopathology images.
 - Providing knowledge guidance for the VLM using a knowledge graph curated by Unified Medical Language System(UMLS) database.
- **Algorithms for Heart Sounds Segmentation** Dec 2024 - Jul 2025
Internship project at The University of Sydney
 - Designed a window-based segmentation pipeline and implemented a stitching algorithm to reconstruction of heart sound phases.
 - Developed a software for hand correction and manual annotation of heart sounds.
- **PPG motion artifact reduction** Sep 2025 - Dec 2025
Implementing algorithms for PPG signal filtering
 - Implemented a hybrid pipeline using signal decomposition and adaptive filtering to mitigate motion artifacts in Photoplethysmogram signals.
 - Evaluated signal quality improvements under various motion-contaminated conditions to validate the pipeline's robustness.
- **Biosense-AI: Deep Learning Based ECG-PPG Signal Classifier and Interpreter** 2023 - 2024
Pocket size electronic device with Mobile app for disease screening
 - Achieving amplified and filtered ECG signal with custom designed PCB only using analog components.
 - SpO₂ and heart sound signals are transferred to OrangePi zero 2W via ADC.
 - Uses the SBC for the processing of signals using multi-modal neural network.
 - Interpretation of possible health issues related to achieved waveforms with SBC and displaying on the mobile-app.

• Robot LUNA: Vision Based Restaurant Robot

2024

Vision based navigating robot with kinect v2 depth camera specifically made for restaurant settings.



- Implementing a vision-based navigation system using Kinectv2 camera with ROS integration for accurate movement.
- Use Gazebo for simulation and comprehensive documentation following the Cambridge EDC model to track design iterations.
- Designing a stabilization tray mechanism to securely transport food items, minimizing spillage with proper calculations.

• Implementing a Fall Detection Device for Eldery People

2023

Device to detect and inform potential falls of eldery people



- Develop algorithm to detect falls of adult individuals using MPU6050 gyroscope by distinguishing from other activities.
- Configured the system to send SMS alerts to a given mobile number after detecting falls through an MQTT server.
- Utilized Altium and SOLIDWORKS for the design and implementation of the prototype.

• Saline Flow Controller

2023

BM2210- Biomedical Device Design.



- Design the prototype for the saline flow controller for maintaining a given constant flow with customizable functions.
- Maintaining proper documentation with the intension of applying a patent and FDA approval.

AWARDS AND SCHOLARSHIPS

• Finalist | Brainstrom

2024

Healthcare competition organized by the IEEE EMBS in University of Moratuwa.

• Semifinalist | Codesprint

2024

*Hackathon organized by IEEE branch of IIT, for **real time sign language translator**.*

• 2nd Runners Up | Mecha

2023

Organized by the Technical Society of Faculty of Medicine in University of Colombo.

• Dialog Merit Scholarship

2021

For the performance in GCE AL examination.

SKILLS

- **Languages:** Python, MATLAB, C, C++, Bash
- **Markup:** HTML, $\text{\LaTeX}$, Markdown
- **Libraries:** Numpy, PyTorch, Scipy, Transformers
- **Tools:** Linux, Git, Docker

VOLUNTEER EXPERIENCE

• IEEE EMBS Student Branch Chapter of University of Moratuwa

2023 - Present

Committee member, Councilor

• Electronics Club, University of Moratuwa

2023 - Present

Member

REFERENCES

1. Dr. Anusha Withana

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The University of Sydney
Email: anusha.withana@sydney.edu.au

2. Dr. Ranga Rodrigo

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